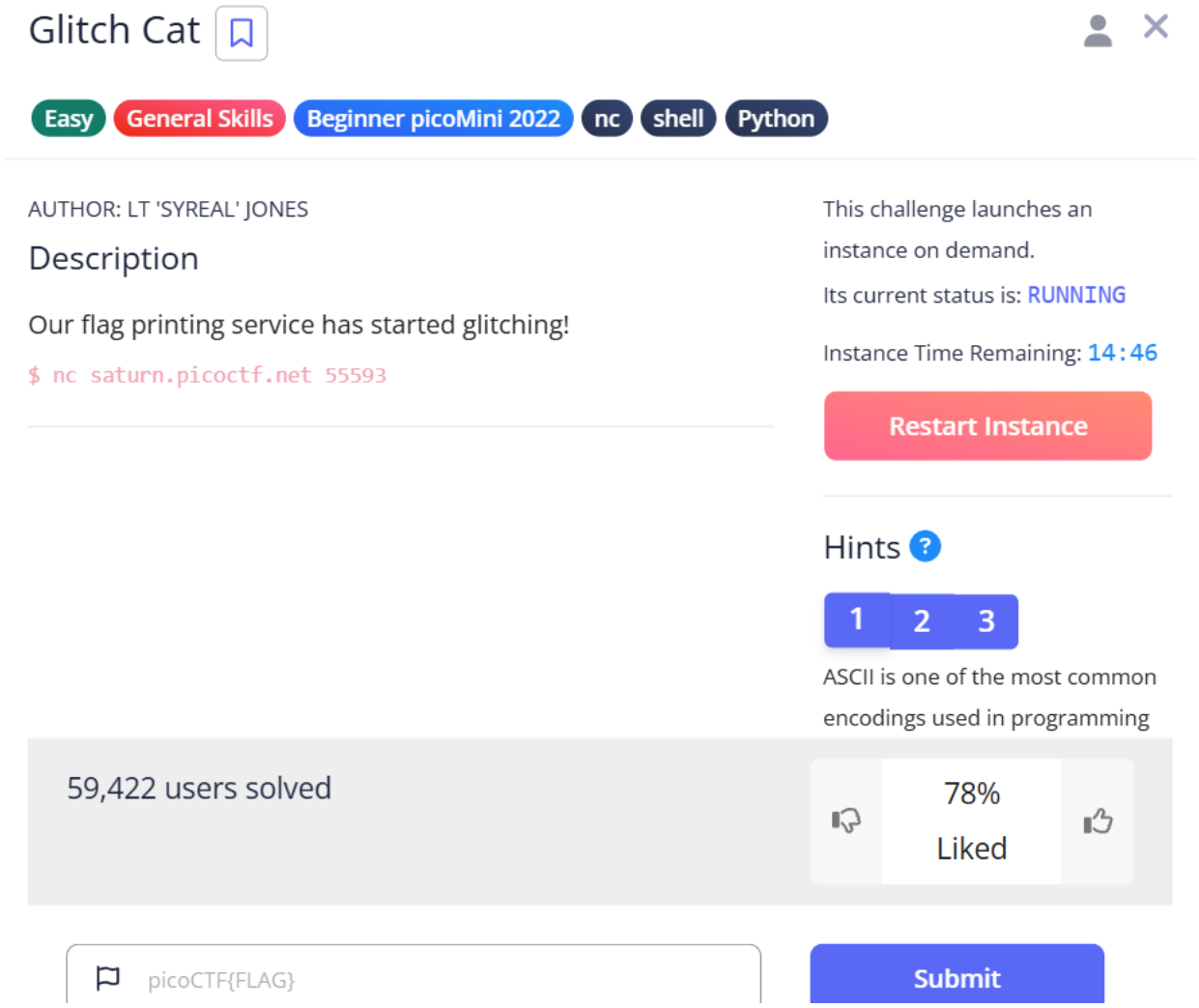
Glitch Cat



Author: The Analyst: Hyposelenia

Challenge: Glitch Cat

Category: General Skills

Date: 01/16/26

I. Objective

The objective of this challenge is to reconstruct the correct picoCTF flag from a "glitched" output by understanding ASCII encoding and recognizing valid Python syntax.

II. Background

In programming, characters are often represented internally using **ASCII values**. Python allows characters to be generated using the chr() function, which converts a number into its corresponding ASCII character.

This challenge presents a partially readable flag combined with ASCII character conversions. By understanding how ASCII encoding works—or by recognizing that the output is valid Python code—the original flag can be reconstructed.

III. Tool Used

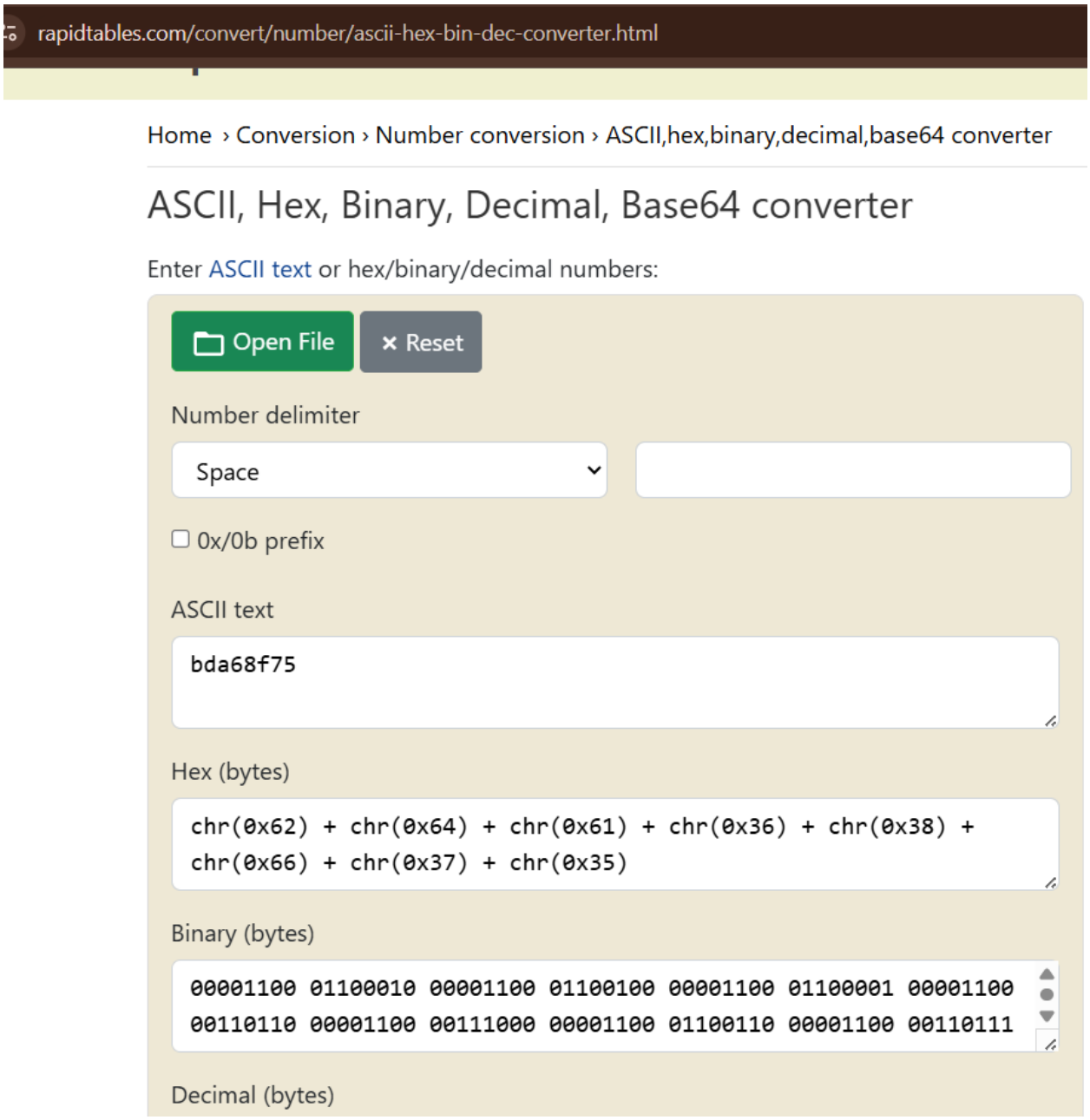
- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **python3** – Used to evaluate Python expressions

IV. Methodology

Method 1: Manual ASCII Conversion

1. Launched the challenge instance and connected to the remote machine.
2. The following output was displayed:
   'picoCTF{gl17ch_m3_n07_' + chr(0x62) + chr(0x64) + chr(0x61) + chr(0x36) + chr(0x38) + chr(0x66) + chr(0x37) + chr(0x35) + '}'
3. Identified the chr(**0x..**) values as **hexadecimal ASCII codes**.

4. Converted each hexadecimal value to its ASCII equivalent using an online hex-to-ASCII converter.



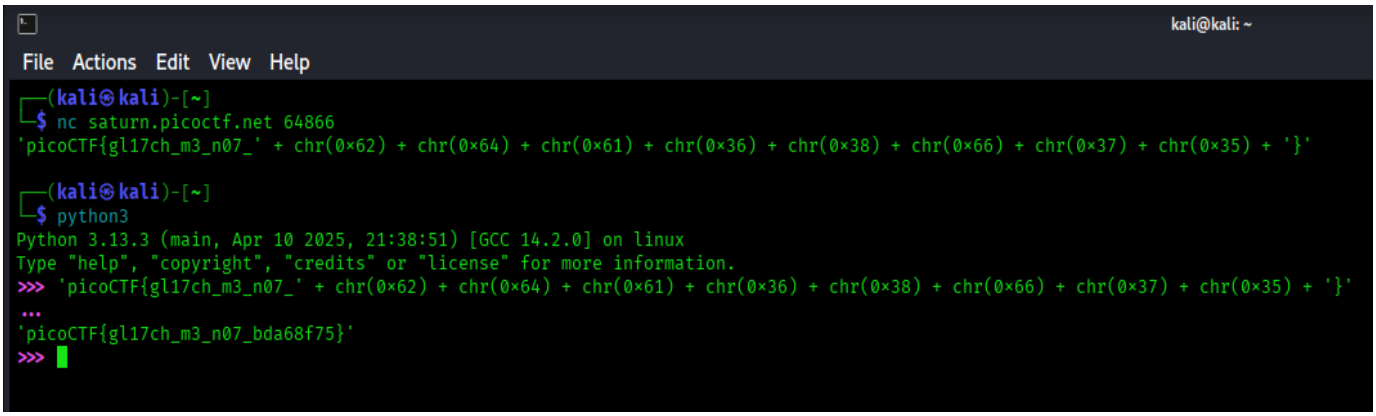
5. Appended the decoded characters to reconstruct the full flag.

Method 2: Executing the Python Expression (More Efficient)

1. Connected to the remote machine and observed the same output.
2. The hint stated that the glitch output is **valid Python**, meaning it can be executed directly.
3. Started the Python interpreter by typing: "***python3***"
4. Pasted the entire glitched output into the Python prompt.
5. Python evaluated the expression and printed the complete flag automatically.

V. Result

picoCTF{gl17ch_m3_n07_bda68f75}



VI. Explanation

This challenge works because Python allows strings to be concatenated using the + operator and characters to be generated using the chr() function. Each chr(0x..) call converts a hexadecimal value into its ASCII character.

- **Method 1** works by manually decoding each character, which helps beginners understand ASCII encoding.
- **Method 2** is more efficient because Python already knows how to interpret the expression. Running it directly lets Python do the decoding for you.

The hint about ASCII explains *what* the values represent, while the hint about valid Python explains *how* to solve the challenge faster.

VII. Conclusion

Glitch Cat teaches two important beginner concepts: ASCII encoding and the power of recognizing valid code syntax. While manual decoding helps build foundational understanding, executing valid Python expressions is faster and more practical in real scenarios. This challenge encourages thinking both logically and creatively when analyzing unusual outputs.

— **The Analyst: Hyposelenia**